

Short Product Information SLC52AIA300A7

Description

- High-Performance, powerful 16-bit contact based security crypto controller with 1 kByte cache.
- SOLID FLASH™ NVM Technology with proven Flash Loader Technology, strong IP protection and EEPROM emulation support.
- Digital Security concept. High speed coprocessor for symmetric and asymmetric cryptography with certified symmetric library (AES and 3-DES) and certified asymmetric library (RSA and ECC). Hybrid-Random Number Generator according to AIS 31 with physical noise source and built-in cryptographic post-processing.
- Memory Management Unit featuring virtual to physical mapping.
- Family concept with the SLC52 products saves development effort.

Features

- 300 kByte SOLID FLASH™
- 12 kByte RAM
- 16-bit architecture based on the MCS® 251 instruction set
- Supported interfaces: UART, I2C, GPIO
- Cryptographic support: HRNG, RSA, ECC, AES, 3-DES
- Standard operating temperature (ambient) -25 to +85°C
- Certification level: CC EAL 6+ high

Applications

- Authentication
- Signature

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2. General Description

SCL52AIA300A7 is part of the SLC52 product series, which is optimized for device authentication applications.

The SLC52AIA300A7 security controller is manufactured in advanced 65nm CMOS technology. It offers easy migration from existing SLE77 and SLE78 product ranges, based on the same 16 bit CPU chip architecture and using the same tool set.

SLC52 features Infineon's SOLID FLASH™ technology. This future-proof memory concept meets growing demand for increasingly differentiated types and levels of functionality in smart cards. It combines the advantages of flexible non-volatile memory (NVM) with a sophisticated security concept. Furthermore, it significantly reduces planning and logistics complexity.

SLC52 is the proven security performance leader. The SLC52 family's proven security concept builds on a high-speed coprocessor supporting both symmetrical cryptography (with AES and DES) and asymmetrical cryptography (with certified RSA and ECC libraries). Chip security is further enhanced by the AIS 31-compliant Hybrid Random Number Generator, which uses a physical noise source and features built-in cryptographic post-processing.

3. Product Overview 3.1

General Features

- 16-bit microcontroller with 24-bit linear addressing
- Highly efficient proprietary architecture based on the 80251 instruction set, with faster execution than a standard 80251 CPU
- Interrupt control unit and Peripheral Event Channel (PEC)
 - Interrupt module for I/O interface and peripherals
 - Enabling fast data transfer through peripheral event channels (PEC)
- **Easy software migration from SLE77 & SLE78**
- **300 kBytes SOLID FLASH™** with free partitioning between code and data
- **12 kBytes RAM**
- **1 kByte unified cache** for optimized code execution
- **Symmetric Crypto Processor (SCP)** for symmetric cryptography (3DES, AES)
- **Crypto engine (Crypto@2304T)** for asymmetric cryptography (RSA and ECC)
- **CRC module** with loadable initialization vector supporting ISO/IEC 3309, CCIT v.41 & HDLC X25
- **Four** 16-bit auto reload **timers**
- **Two Watchdog** timers (WDT)
- External clock frequency: 1 to 10 MHz
- **Internal clock frequency up to 85 MHz**
- **Adjustable internal frequency acc. to available power or required performance**
 - Increased internal frequency for maximum Performance
 - Internal frequency is automatically adjusted to maintain a given power limit
- Supply voltage range: 1.62 V to 5.5 V
- Support of current consumption limits: <10mA @5.5V; <6mA @3.3V; <4mA @1.98V
- Operating temperature range: -25°C ... +85°C

- Power-saving sleep mode

3.2 Contact-based Interface

- **I2C** slave in standard mode (up to 100 kbit/s), fast mode (up to 400 kbit/s) and fast mode+ (up to 1000 kbit/s)
- **Enhanced UART** for handling serial interface in accordance with ISO/IEC 7816 part 3 supporting transmission protocols T=0 and T=1 (support of clock division factor of 8)

3.3 SLC52A Platform Security Features

Digital Security

- Digital Security concentrates on digital, mathematically-modeled security mechanisms and comprehensive error detection over the complete data path
- Error Detection Codes (EDC) in all memories
- EDC for cache protection

Main security features

- Crypto engine (Crypto@2304T) for public key cryptography
- RSA calculations with key lengths of up to:
 - 2048 bits w/o CRT
 - 4096 bits with CRT
- Elliptic Curve Cryptography (ECC) with key lengths up to 521 bits
- Support of ECDSA (Elliptic Curves Digital Signature Algorithm) and ECDH (Elliptic Curve Diffie-Hellman) in prime fields GF(p) as well as binary fields GF(2ⁿ) for key lengths up to 521 bits
- Symmetric Crypto Processor (SCP) for triple-DES and AES acceleration
- Robust set of threshold sensors and filters for light, temperature, frequency and voltage
- Hybrid random number generator (HRNG) – fulfilling the quality criteria of the BSI AIS20/31 for the functionality classes DRG.2, DRG.3, PTG.2 and PTG.3. The HRNG consists of a PTG.2 compliant True Random Number Generator (TRNG), and a DRG.3 compliant Pseudo/Deterministic Random

- Special Function Register (SFR) encryption
- User mode Security Life Control (UmSLC)
- Two Watchdog timers (WDT) incl. checkpoint register for runtime checks
- High-security Self-Test Software concept
- Security reset mechanism
- Effective countermeasures against SEMA/DEMA, SPA/DPA, DFA and timing-attacks

Memory Security

- **Memory Management and Protection Unit (MMU)** with 8 levels of hardware-supported firewalls of up to 16 MBytes linear address space on each level
- MED – Memory Encryption/Decryption device for RAM, IFX-ROM and NVM in code and data areas
- Unique chip identification number for each chip

3.4 Certification level

- CC EAL6+ (High)¹ according to Common Criteria 3.1 and protection profile PP0084

3.5 SOLID FLASH™ Technology

- Based on reliable NVM SOLID FLASH™ Technology
- Protection of content by strong encryption in addition to error detection and single-bit correction
- **Certified SOLID FLASH™ loader concept:**
 - High-speed flash loading for fast personalization
 - Optional **SOLID FLASH™** programming service by Infineon
- Certified Hardware Support Library (HSL) for flexible and reliable Flash programming:
 - EEPROM-Emulation Support using In-Place-Update for easy software migration

- Performance boost for Operating Systems Utilizing Incremental Write

Minimum 1.6 million block write

Operations per page (8 blocks per page)

- Typical data retention of >10 years @ 25 °C
- Flash programming voltage generated on-chip
- Flash programming voltage generated on-chip
- ESD robustness
 - Human body model (HBM) 4 kV
 - Charged device model (CDM) 750 V

3.6 Development Tools Overview

- Software Development Kit, SDK 70 (based on the Keil DK251 kit)
- FPGA-based emulator ET70
- Certified symmetric and asymmetric libraries
- Application Notes
- Worldwide application engineer team & dedicated field application engineers
- Regular trainings on hardware & software tools
- On-site trainings available on request

¹For more information see: www.bsi.bund.de

4. Document References

- Confidential Hardware Reference Manual (HRM)
- Confidential Programmers Reference Manual (PRM)
- Confidential Product List
- Confidential Security Guidance
- Confidential Chip qualification report
- Confidential Module specification containing package descriptions

5. Performance

The internal clock frequency can be adjusted to a level of up to 85 MHz. It is adjustable according to either available power requirements or required performance:

- Increased internal clock frequency for maximum performance, e.g. for high performance at maximum frequency
- Automatically adjusted frequency to maintain a given maximum power consumption

6. Peripherals

To enable high performance cryptography, the SLC52 controllers provide two cryptographic coprocessors: the Crypto@2304T for modular arithmetic and asymmetric cryptographic algorithms, plus the Symmetric Crypto Processor (SCP) for symmetric cryptographic algorithms.

6.1 Crypto@2304T

The Crypto@2304T is a high-security and high-performance crypto coprocessor. It is equipped with its own RAM of 1152 bytes and supports all of today's known public key algorithms based on large integer modular arithmetic with operands with register lengths of up to 2304 bits. It allows fast and efficient calculation of arithmetic operations, such as RSA and ECC. The features of the Crypto@2304 are supported by the RSA2048, RSA4096 and ECC521 (ECDSA, ECDH) libraries.

6.2 SCP

The SCP (Symmetric Crypto Processor) module supports symmetrical crypto algorithms according to the Data Encryption Standard (DES) in the Electronic Code Book Mode, as well as Cipher Block Chaining. The AES (Advanced Encryption Standard) component of the SCP module performs AES-compliant operations for all three different key lengths (128, 192, 256 bits).

6.3 ICU and PEC

The Interrupt Control Unit processes interrupt requests from different sources to run an interrupt service routine (ISR). Data can be directly transferred between memory locations with minimal CPU activity for fast

Interaction with peripherals by using so-called Peripheral Event Channels (PECs). The channels can be assigned individually to peripherals or chained together to enable continuous transfer without handover delay between the channels. This is of particular advantage when power-aware programming is required.

6.4 CRC

The enhanced CRC module allows easy generation of checksums according to ISO/IEC 3309 (16-bit-CRC) and offers a loadable initialization vector for optimized Java support.

6.5 Contact based interface

I2C slave in standard mode (up to 100kbit/s), fast mode (up to 400kbit/s) and fast mode+ (up to 1000 kbit/s)

Enhanced Universal Asynchronous Receiver/Transmitter Interface (UART) for handling serial interface in accordance with ISO/IEC 7816 part 3 supporting transmission protocols T=0 and T=1 (support of clock division factor of 8)

6.6 Timers and Watchdogs

The four integrated 16-bit auto-reload timers support easy implementation of advanced communication protocols (T=0 and T=1) and all other critical timing processes. The timers have the same functionality and support the same features. The two Watchdog timers (WDT) is a circuit that monitors controller operation by automatically initiating a security reset if a specified period without an adequate response elapses after occurrence of a hardware or software irregularity.

6.7 RNG

The advantages of the two random number generation (RNG) types, physical true random number generation (PTRNG) and deterministic random number generation (DRNG), are combined in a hybrid RNG (HRNG) on this chip.

The HRNG provides random number generation as a security service to the user and for internal purposes. The genuine random numbers produced can be used directly for any application demanding excellent physical random data quality. The Hybrid random number generator (HRNG) fulfills the quality criteria of the BSI AIS20/31 for the functionality classes DRG.2, DRG.3, PTG.2 and PTG.3.

6.8 Power Management Module

The SLC52 controllers' power management module offers advanced specific power-saving modes. In the halt mode the CPU clock is stopped, but the clocks serving the peripheral units continue running. In the sleep mode the chip's oscillator stops and the controller remains in a static state. The chip is returned to the normal operating mode in response to an interrupt or reset.

7. Memory & Block Diagram

The SCL52AIA300A7 offers 12 kBytes of RAM and 300 kBytes of SOLID FLASH™ to meet the demanding requirements of device authentication applications.

The block diagram shows the basic functional components of the controller and their interconnection. Code and data are stored both in a linear memory space, allowing direct access without the need to swap memory segments.

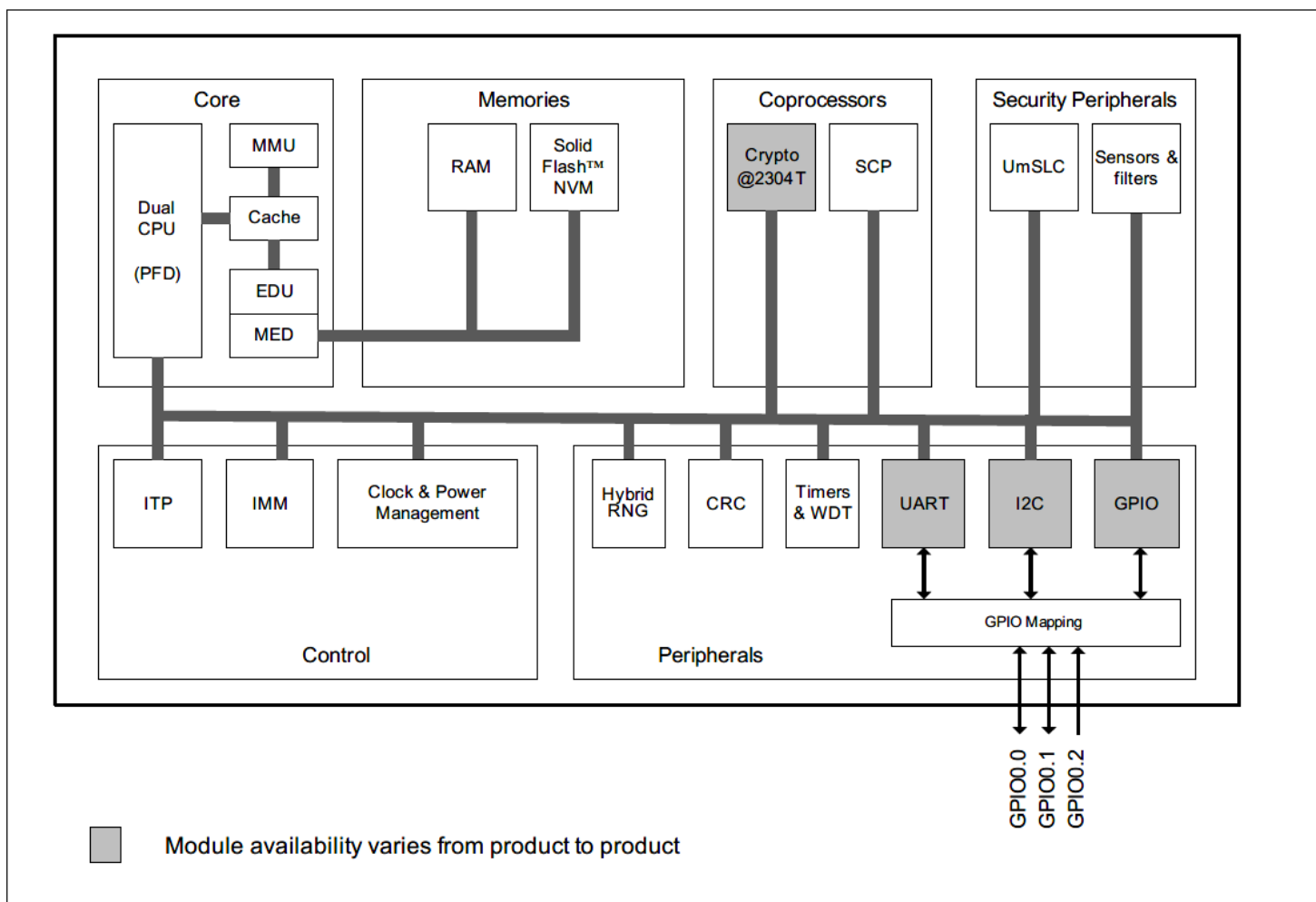


Figure 1: Block Diagram of the SCL52AIA300A7

8. Ordering Information

These are the main "ordering sales codes" for SCL52AIA300 Product.

Ordering sales codes	Package	Common features
SLC52AIA300A7 USON10	USON 10	- NVM: 300KB; RAM 12KB - Interfaces: I2C slave mode, UART, GPIO, ISO/IEC 7816 - Crypto acceleration: AES/DES, RSA/ECC, SHA-256 - Certification level: Common Criteria EAL 6+ (high) acc. to protection profile PP0084 - System frequency (CPU clock): up to 85 MHz - Voltage range: 1.62 V to 5.5 V

Table 1: Ordering Sales Codes with package product information

Flash initialization/personalization and additional delivery forms (ordering codes) available upon request. For ordering information please refer to the Hardware Reference Manual and contact your sales representative.

Note: It is recommended to prevent exposure to intense light during chip operation.

9. Description of Package

This chapter provides information about available delivery forms and how the product's interfaces are assigned to the package pins.

For each package or module, a specification containing information about geometry, materials, delivery forms, etc. can be ordered from Infineon Technologies AG at dsscustomerservice@infineon.com

Note: The drawings below are for information only and not drawn to scale. More detailed information about package characteristics and assembly instructions is available on request.

9.1 SMD packages

The following package is available:

- PG-USON-10

The PG-USON-10 package layout and the pin-to-signal reference care shown below.

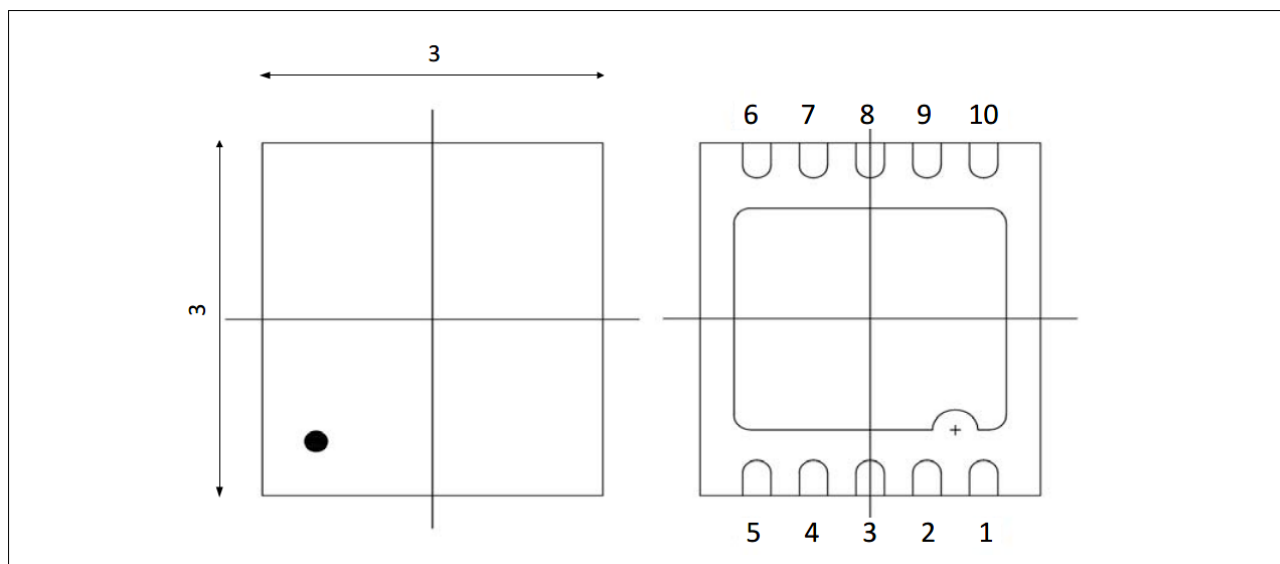


Figure 2 Configuration for PG-USON-10 (all dimensions in mm)

Table Pin-to-Signal Reference for PG-USON-10

Pin	Symbol	Direction	Pad Type	Signal Function / Remarks
01	GND	(n.a.)	Power supply	Common ground reference (V_{SS})
02	NC			No internal connection / do not connect externally
03	GPIO0.1	I/O	GPIO	Software I/O (SWIO) usage: GPIO0.1 I2C usage (all modes): SDA ISO/IEC 7816-3 card usage: UART_IO
04	NC			No internal connection / do not connect externally
05	NC			No internal connection / do not connect externally

Pin	Symbol	Direction	Pad Type	Signal Function / Remarks
06	NC			No internal connection / do not connect externally
07	NC			No internal connection / do not connect externally
08	GPIO0.0	I/O	GPIO	Software I/O (SWIO) usage: GPIO0.0 I2C usage (normal mode): SCL ISO/IEC 7816-3 card usage: UART_CLK
09	GPIO0.2	I	GPIO	Software I/O (SWIO) usage: GPIO0.2 I2C usage (normal mode): SCL ISO/IEC 7816-3 card usage: UART_RST
10	V _{CC}	(n.a.)	Power supply	Power and pad supply (V _{CC})

10. Sales Code Description

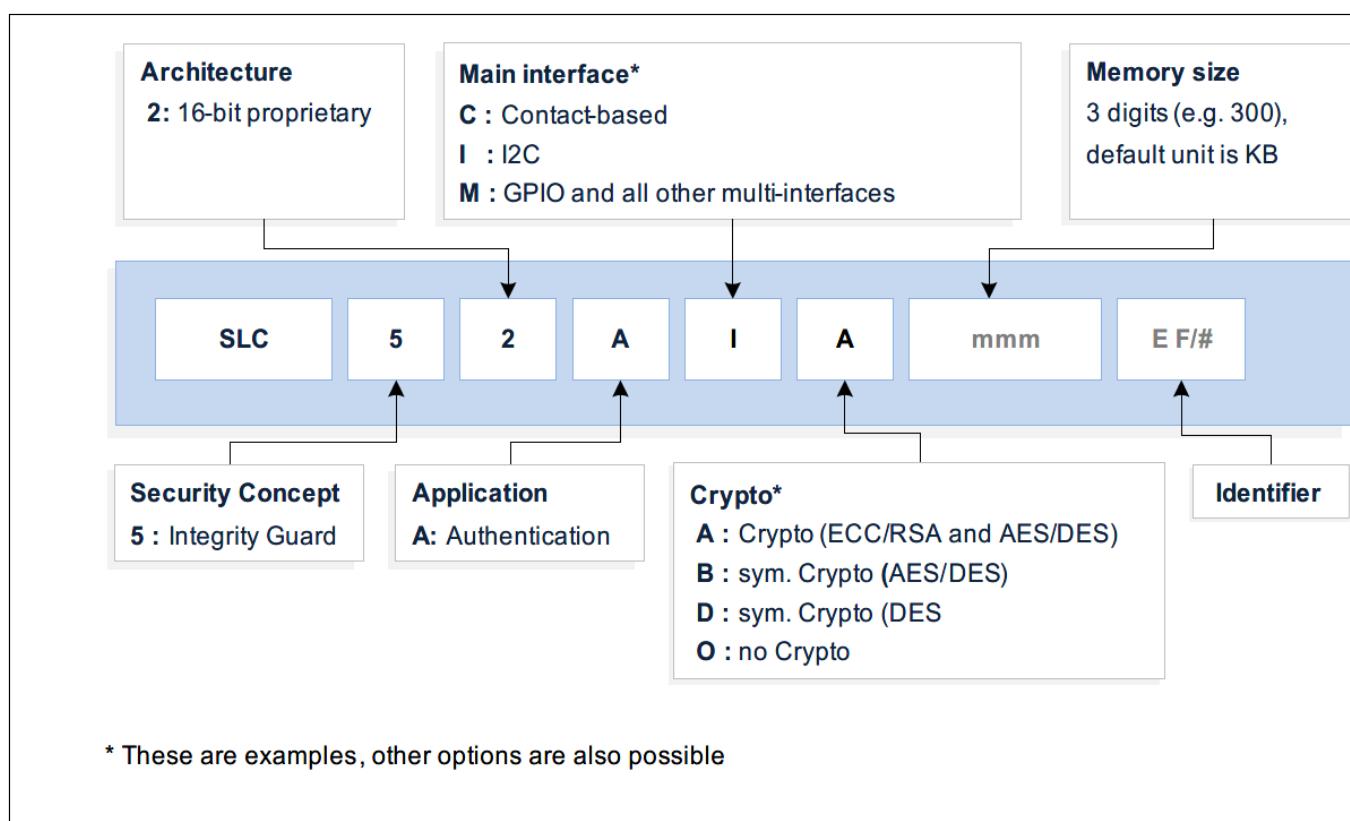


Figure 5: Sales Code Description

11. Glossary

		ID	Identity
AES	Advanced Encryption Standard	I/O	Input/Output
CLK	Clock	ISO/IEC	International Organization for Standardization / International Electrotechnical Commission
CRC	Cyclic Redundancy Check		
CRT	Chinese Remainder Theorem		
CPU	Central Processing Unit		
CMOS	Complementary Metal-Oxide Semiconductor	IMM	Interface Management Module
DES	Data Encryption Standard	ISR	Interrupt Service Routine
DFA	Differential Fault Analysis	MED	Memory Encryption Decryption unit
DPA	Differential Power Analysis	MMU	Memory Management Unit
DEMA	Differential Electromagnetic Analysis	NVM	Non Volatile Memory
ECC	Elliptic Curve Cryptography	OS	Operating System
ECDH	Elliptic Curve Diffie Hellman	PEC	Peripheral Event Channel
ECDSA	Elliptic Curve Digital Signature Algorithm	TRNG	Pseudo Random Number Generator
EDC	Error Detection Code	RAM	Random Access Memory
EMV	Europa Mastercard Visa Co compliance testing	ROM	Read-Only Memory
E ² PROM	Electrically Erasable Programmable Read-Only Memory (equivalent to NVM)	RSA	Rivest Shamir Adleman algorithm
ESD	Electrostatic Discharge, release of static electricity that can damage a chip	RST	Reset
ETSI	European Telecommunication Standards Institute	SCP	Symmetric Crypto Processor
FIFO	First In, First Out	SDK CC	Software Development Kit Chip Card
FPGA	Field Programmable Gate Array	SEMA	Simple Electromagnetic Analysis
GND	Ground	SFR	Special Function Register
GPIO	General Purpose Input Output	SPA	Simple Power Analysis
HBM	Human Body Model	STS	Self-Test Software
		T=0, T=1	Communication Protocols defined in ISO 7816 standard
		TRNG	True Random Number Generator
		UART	Universal Asynchronous Receiver/Transmitter
		WDT	Watchdog timer

12. Revision History

These are the major changes since the last revision.

Page or Reference	Description of change
Revision 1.0, 2019-09-19	First released version

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